PRÁCTICA: KUBERNETES CON LOAD BALANCER Y AWS

Introducción

- **Kubernetes** es un sistema que gestiona automáticamente aplicaciones en múltiples máquinas. Sirve para desplegar, escalar, supervisar y recuperar aplicaciones sin intervención manual.
  - **k3d** es una versión ligera de Kubernetes (*50 MB vs 500 MB*). Funciona increíblemente bien en WSL2 sin Docker ni complicaciones.
- **POD** es la unidad mínima de Kubernetes, un contenedor ejecutándose. Tiene IP propia, es efímero (*desaparece cuando muere*), aislado.
- **Deployment** es un controlador que crea y mantiene múltiples pods idénticos, su responsabilidad es que, si un pod muere, crea uno nuevo. Mantiene la cantidad especificada.

• **Service** es un intermediario que da una IP estable para acceder a pods. Los pods tienen IPs que cambian. Service proporciona una IP que nunca cambia. • **Load Balancer** es un tipo de Service que distribuye solicitudes entre múltiples pods. El LoadBalancer elige qué Pod atender (*round robin, menos conexiones, etc.*) va a ser atendido el cliente. Si un pod cae, otros atienden. La carga se distribuye.

• **Namespace** es un espacio aislado dentro de Kubernetes (*como carpetas*). Separar desarrollo/producción, equipos, versiones diferentes. • **ConfigMap** almacena configuración (*URLs, puertos, etc.*). Cambias config sin recompilar ni redeploy.

- **Secret** es un ConfigMap pero para datos sensibles (*contraseñas, tokens*). Está cifrado, no es visible en logs.
- **Labels** son etiquetas para identificar objetos (*app: web-app, version: 1.0*). • **Selectors** son formas de filtrar objetos por sus labels.

• **Escalado horizontal** es aumentar número de pods (*de 3 a 5*). La carga se distribuye entre más máquinas, Kubernetes lo hace automáticamente. • **Escalado vertical** es aumentar recursos de una máquina (2GB → 8GB RAM), tiene límites.

• **Session Affinity** mantiene al cliente en el mismo pod si ya está conectado. • **Port-forward** es un túnel que expone puertos de Kubernetes en tu máquina local. Su uso es solo para desarrollo/testing.

- **Túnel SSH** es una conexión SSH que redirige puertos desde máquina remota a local. EC2 accede a Kubernetes sin ruta de red directa.

• **Ingress** es un objeto que gestiona acceso externo a servicios (*dominios, HTTPS, routing*). Su ventaja sobre port-forward es que tiene nombres reales (*miapp.com*), HTTPS, y su uso es profesional.
• **Persistent Volume** es almacenamiento que persiste más allá de la vida del pod. Si un pod muere, sus datos desaparecen. PV los guarda en almacenamiento externo.
• **StateFulset** es como Deployment pero para aplicaciones con estado (*BD, colas, etc.*). Los pods tienen identidad. El orden importa.
• **Job** ejecuta una tarea UNA VEZ y termina (*no indefinido como Deployment*). Se usa en migraciones, backups, procesamiento por lotes.
• **CronJob** es un Job que se ejecuta en un horario específico. • **Yaml** es el formato para describir objetos de Kubernetes de forma legible.

Arquitectura Simple

1. CLIENTE
2. PORT-FORWARD / INGRESS
3. SERVICE (*IP estable*)
4. LOAD BALANCER (*elige pod*)
5. DEPLOYMENT (*3 replicas*)
6. PODS (*tu aplicación*)

Flujo de una Petición

1. Cliente solicita http://localhost:8080
2. Port-forward escucha, redirige a Service
3. Service elige qué pod (*LoadBalancer*)
4. Pod ejecuta tu código (*Flask, Node, Java, etc.*)
5. Respuesta vuelve al cliente

Si un pod muere, Deployment crea uno nuevo automáticamente.

Objetivos

Al finalizar esta práctica, serás capaz de:

• Instalar Kubernetes local directamente en WSL2 (*sin Docker, sin dependencias*)

• Desplegar múltiples replicas de una aplicación en Kubernetes

• Configurar un Load Balancer interno para distribuir tráfico •

Conectar tu cluster local de Kubernetes con instancias EC2 en AWS

• Monitorear el tráfico y verificar el balanceo de carga • Escalar

aplicaciones dinámicamente en Kubernetes

Requisitos Previos

• Windows 11 con WSL2 habilitado

• Ubuntu 22.04 o superior en WSL2

• kubectl instalado

• k3s instalado (*Kubernetes ligero sin Docker*)

• Cuenta AWS Free Tier

• Acceso SSH a instancias EC2

• Terminal en Windows o WSL

• Al menos 2 GB RAM disponibles

Requisitos de AWS (*100% alcanzable con Free*

***Tier*)** • Acceso a EC2 (*crear/gestionar instancias*)

• Acceso a Security Groups

• Acceso a Key Pairs

PARTE 0: PREPARACIÓN DEL ENTORNO LOCAL

EN WSL2 0.1 – Instalar k3s (Kubernetes ultraligero - SIN

Docker)

k3s es Kubernetes completo, pero sin la complejidad. Es perfecto para desarrollo y testing. En WSL2 (*Ubuntu*):

 # Actualizar sistema

sudo apt update – y && sudo apt upgrade -y

```
root@ubunturuben:/home/izan# apt update && apt upgrade -y
Get:1 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version InRelease
.8 kB]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://es.archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 14.8 kB in 1s (26.4 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
8 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following upgrades have been deferred due to phasing:
  gir1.2-mutter-14 gir1.2-nm-1.0 libmutter-14-0 libnm0 mutter-common
  mutter-common-bin network-manager network-manager-config-connectivity-ubuntu
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
```

 # Instalar dependencias mínimas

sudo apt install -y curl wget git

```
root@ubunturuben:/home/izan# sudo apt install -y curl wget git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following package was automatically installed and is no longer required:
  libllvm19
```

 # Instalar k3s SIN systemd (mejor para WSL2)

curl -sfL https://get.k3s.io | K3S_KUBECONFIG_MODE="644" sh -

```
root@ubunturuben:/home/izan# curl -sfL https://get.k3s.io | K3S_KUBECONFIG_MODE="644" s
h -
[INFO]  Finding release for channel stable
[INFO]  Using v1.34.3+k3s1 as release
[INFO]  Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.34.3+k3s1/s
ha256sum-amd64.txt
[INFO]  Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.34.3+k3s1
/k3s
```

 # Verificar que k3s está instalado

sudo k3s --version

```
root@ubunturuben:/home/izan# sudo k3s --version
k3s version v1.34.3+k3s1 (48ffa7b6)
go version go1.24.11
```

 # Iniciar k3s y esperar 10 segundos a que inicie

sudo k3s server & sleep 10

```
root@ubunturuben:/home/izan# sudo k3s server & sleep 10
[1] 5908
INFO[0000] Starting k3s v1.34.3+k3s1 (48ffa7b6)
INFO[0000] Configuring sqlite3 database connection pooling: maxIdleConns=2, maxOpenConn
s=0, connMaxLifetime=0s
INFO[0000] Configuring database table schema and indexes, this may take a moment...
INFO[0000] Database tables and indexes are up to date
INFO[0000] Kine available at unix://kine.sock
INFO[0000] Reconciling bootstrap data between datastore and disk
```

 # Verificar que está corriendo

sudo k3s kubectl get nodes

```
root@ubunturuben:/home/izan# sudo k3s kubectl get nodes
NAME          STATUS   ROLES           AGE   VERSION
ubunturuben   Ready    control-plane   32s   v1.34.3+k3s1
root@ubunturuben:/home/izan#
```

0.2 – Instalar kubectl localmente (*para comodidad*)

 # Descargar kubectl

curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

```
root@ubunturuben:/home/izan# curl -LO "https://dl.k8s.io/release/$(curl -L -s
>
> https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
curl: (2) no URL specified
curl: try 'curl --help' or 'curl --manual' for more information
bash: https://dl.k8s.io/release/stable.txt: No such file or directory
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   138  100   138    0     0    629      0 --:--:-- --:--:-- --:--:--   630
100   238  100   238    0     0    515      0 --:--:-- --:--:-- --:--:--   515
root@ubunturuben:/home/izan#
```

sudo chmod +x kubectl

sudo mv kubectl /usr/local/bin/

```
root@ubunturuben:/home/izan# sudo chmod +x kubectl
root@ubunturuben:/home/izan# sudo mv kubectl /usr/local/bin/
```

 # Verificar

kubectl version --client

```
/usr/local/bin/kubectl: line 1: `<?xml version='1.0' encoding='UTF-8'?><Error><Code>NoS
uchKey</Code><Message>The specified key does not exist.</Message><Details>No such objec
t: 767373bbdcb8270361b96548387bf2a9ad0d48758c35/release/bin/linux/amd64/kubectl</Detail
s></Error>'
root@ubunturuben:/home/izan#
```

 # Configurar kubeconfig para acceder sin sudo

mkdir -p $HOME/.kube

```
root@ubunturuben:/home/izan# mkdir -p $HOME/.kube
```

sudo cp /etc/rancher/k3s/k3s.yaml $HOME/.kube/config

sudo chown $(id -u):$(id -g) $HOME/.kube/config

sudo chmod 600 $HOME/.kube/config
 # Verificar que funciona sin sudo

```
root@ubunturuben:/home/izan# mkdir -p $HOME/.kube
root@ubunturuben:/home/izan# sudo cp /etc/rancher/k3s/k3s.yaml $HOME/.kube/config
root@ubunturuben:/home/izan# sudo chown $(id -u):$(id -g) $HOME/.kube/config
root@ubunturuben:/home/izan# sudo chmod 600 $HOME/.kube/config
root@ubunturuben:/home/izan#
```

kubectl get nodes

```
/usr/local/bin/kubectl: line 1: `<?xml version='1.0' encoding='UTF-8'?><Error><Code>NoS
uchKey</Code><Message>The specified key does not exist.</Message><Details>No such objec
t: 767373bbdcb8270361b96548387bf2a9ad0d48758c35/release/bin/linux/amd64/kubectl</Detail
s></Error>'
```

0.3 – Crear directorio de trabajo

mkdir -p ~/kubernetes-aws-practice

SSH

```
root@ubunturuben:/home/izan# mkdir -p ~/kubernetes-aws-practice
root@ubunturuben:/home/izan# cd ~/kubernetes-aws-practice
root@ubunturuben:~/kubernetes-aws-practice#
```

PARTE 1: CREAR APLICACIÓN SIMPLE PARA KUBERNETES 1.1 – Crear carpeta para la aplicación

mkdir -p ~/kubernetes-aws-practice/app

cd ~/kubernetes-aws-practice/app

```
root@ubunturuben:~/kubernetes-aws-practice# mkdir -p ~/kubernetes-aws-practice/app
root@ubunturuben:~/kubernetes-aws-practice# cd ~/kubernetes-aws-practice/app
root@ubunturuben:~/kubernetes-aws-practice/app#
```

1.2 – Crear aplicación Python con Flask

Crear app.py:

```
cat > app.py << 'EOF'

#!/usr/bin/env python3

from flask import Flask, jsonify, send_from_directory

import os

import socket

from datetime import datetime

import sys

app = Flask(__name__)

# Variables de entorno inyectadas por Kubernetes

POD_NAME = os.getenv('POD_NAME', 'Unknown Pod')

POD_NAMESPACE = os.getenv('POD_NAMESPACE', 'default')

@app.route('/')

def index():

 return send_from_directory('.', 'index.html')

@app.route('/pod-info')
def pod_info():

 return jsonify({

 'pod_name': POD_NAME,

 'namespace': POD_NAMESPACE,

 'hostname': socket.gethostname(),

 'timestamp': datetime.now().isoformat()

 })

@app.route('/health')

def health():

 return jsonify({'status': 'healthy', 'pod': POD_NAME}),

200 if __name__ == '__main__':

 print(f"[{POD_NAME}] Iniciando servidor Flask...",

file=sys.stderr)  app.run(host='0.0.0.0', port=5000,

debug=False) EOF
```

sudo chmod +x app.py

```
root@ubunturuben:~/kubernetes-aws-practice/app# cat > app.py << 'EOF'
#!/usr/bin/env python3
from flask import Flask, jsonify, send_from_directory
import os
import socket
from datetime import datetime
import sys
app = Flask(__name__)
# Variables de entorno inyectadas por Kubernetes
POD_NAME = os.getenv('POD_NAME', 'Unknown Pod')
POD_NAMESPACE = os.getenv('POD_NAMESPACE', 'default')
@app.route('/')
def index():
 return send_from_directory('.', 'index.html')
@app.route('/pod-info')
def pod_info():
 return jsonify({
 'pod_name': POD_NAME,
 'namespace': POD_NAMESPACE,
 'hostname': socket.gethostname(),
 'timestamp': datetime.now().isoformat()
 })
@app.route('/health')
def health():
 return jsonify({'status': 'healthy', 'pod': POD_NAME}), 200 if __name__ == '__main__':

 print(f"[{POD_NAME}] Iniciando servidor Flask...", file=sys.stderr)  app.run(host='0.0
.0.0', port=5000, debug=False) EOF
> sudo chmod +x app.py
> S
```

1.3 – Crear archivo HTML

cat > index.html << 'EOF'

<!DOCTYPE html>

<html>

<head>

<title>Kubernetes Load Balancer</title>

<style>

body {

font-family: Arial, sans-serif;

display: flex;

justify-content: center;

align-items: center;

```
min-height: 100vh;
margin: 0;
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  }
.container {
background: white;
padding: 50px;
border-radius: 10px;
box-shadow: 0 10px 25px rgba(0,0,0,0.2);
text-align: center;
max-width: 500px;
}
h1 {
color: #667eea;
margin: 0 0 30px 0;
}
.info {
background: #f0f0f0;
padding: 20px;
border-radius: 5px;
margin: 20px 0;
}
.pod-name {
font-size: 28px;
color: #764ba2;
font-weight: bold;
font-family: monospace;
}
.timestamp {
color: #666;
font-size: 13px;
margin-top: 10px;
```

```
}
.instruction {
background: #e0f2fe;
padding: 15px;
border-radius: 5px;
color: #0369a1;
font-size: 14px;
margin-top: 20px;
}
.refresh-button {
margin-top: 20px;
padding: 10px 20px;
background: #667eea;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-size: 14px;
}
.refresh-button:hover {
background: #764ba2;
}
</style>
</head>
<body>
<div class="container">
<h1> Kubernetes Load Balancer</h1>
<div class="info">
<p style="margin: 0 0 10px 0; color: #666;">Pod atendiendo solicitud:</p>
<p class="pod-name" id="pod-name">Cargando...</p>
<p class="timestamp" id="timestamp"></p>
```

```
</div>
<div class="instruction">
Actualiza constantemente esta página para ver el <b>balanceo de carga en acción</b>
</div>
<button class="refresh-button" onclick="location.reload()">Actualizar Ahora</button>
</div>
<script>
fetch('/pod-info')
.then(r => r.json())
.then(data => {
document.getElementById('pod-name').textContent = data.pod_name;
const date = new Date(data.timestamp);
document.getElementById('timestamp').textContent =
date.toLocaleString('es-ES');
})
.catch(e => {
document.getElementById('pod-name').textContent = ' Error';
console.error('Error:', e);
});
</script>
</body>
</html>
EOF
```

```
<html>
<head>
 <title>Kubernetes Load Balancer</title>
 <style>
 body {
 font-family: Arial, sans-serif;
 display: flex;
 justify-content: center;
 align-items: center;
 min-height: 100vh;
 margin: 0;
 background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  }
 .container {
 background: white;
 padding: 50px;
 border-radius: 10px;
 box-shadow: 0 10px 25px rgba(0,0,0,0.2);
 text-align: center;
 max-width: 500px;
 }
 h1 {
 color: #667eea;
 margin: 0 0 30px 0;
 }
 .info {
 background: #f0f0f0;
 padding: 20px;
 border-radius: 5px;
 margin: 20px 0;
 }
 .pod-name {
 font-size: 28px;
 color: #764ba2;
 font-weight: bold;
 font-family: monospace;
 }
 .timestamp {
EOF ml> t> etElementById('pod-name').textContent = ' Error';  console.error('Error:', e
> 
```

1.4 – Crear requirements.txt

cat > requirements.txt << 'EOF'

Flask==3.0.0

Werkzeug==3.0.0

EOF

```
> cat > requirements.txt << 'EOF'
> Flask==3.0.0
Werkzeug==3.0.0
EOF
root@ubunturuben:~/kubernetes-aws-practice/app#
```

1.5 – Crear Dockerfile ultraligero

Nota, este Dockerfile es solo para construcción local. k3s lo ejecutará directamente:

cat > Dockerfile << 'EOF'

FROM python:3.11-slim

WORKDIR /app

 # Copiar archivos

COPY requirements.txt .

COPY app.py .

COPY index.html .

 # Instalar dependencias

RUN pip install --no-cache-dir -r requirements.txt

 # Ejecutar app

CMD ["python", "app.py"]

EOF

```
root@ubunturuben:~/kubernetes-aws-practice/app# cat > Dockerfile << 'EOF'
FROM python:3.11-slim
WORKDIR /app
# Copiar archivos
COPY requirements.txt .
COPY app.py .
COPY index.html .
# Instalar dependencias
RUN pip install --no-cache-dir -r requirements.txt
# Ejecutar app
CMD ["python", "app.py"]
EOF
```

PARTE 2: CONFIGURAR KUBERNETES (*MANIFESTS YAML*) 2.1 – Crear Namespace

cd ~/kubernetes-aws-practice

cat > namespace.yaml << 'EOF'

apiVersion: v1

kind: Namespace

metadata:

name: load-balancer-demo

labels:

name: load-balancer-demo

EOF
 # Aplicar

kubectl apply -f namespace.yaml

 # Verificar

kubectl get namespaces

```
root@ubunturuben:~/kubernetes-aws-practice/app# cat > Dockerfile << 'EOF'
FROM python:3.11-slim
WORKDIR /app
# Copiar archivos
COPY requirements.txt .
COPY app.py .
COPY index.html .
# Instalar dependencias
RUN pip install --no-cache-dir -r requirements.txt
# Ejecutar app
CMD ["python", "app.py"]
EOF
> cd ~/kubernetes-aws-practice
cat > namespace.yaml << 'EOF'
apiVersion: v1
kind: Namespace
metadata:
 name: load-balancer-demo
 labels:
 name: load-balancer-demo
EOF
# Aplicar
kubectl apply -f namespace.yaml
# Verificar
kubectl get namespaces
/usr/local/bin/kubectl: line 1: syntax error near unexpected token `<'
/usr/local/bin/kubectl: line 1: `<?xml version='1.0' encoding='UTF-8'?><Error><Code>NoS
uchKey</Code><Message>The specified key does not exist.</Message><Details>No such objec
t: 767373bbdcb8270361b96548387bf2a9ad0d48758c35/release/bin/linux/amd64/kubectl</Detail
s></Error>'
/usr/local/bin/kubectl: line 1: syntax error near unexpected token `<'
/usr/local/bin/kubectl: line 1: `<?xml version='1.0' encoding='UTF-8'?><Error><Code>NoS
uchKey</Code><Message>The specified key does not exist.</Message><Details>No such objec
t: 767373bbdcb8270361b96548387bf2a9ad0d48758c35/release/bin/linux/amd64/kubectl</Detail
s></Error>'
root@ubunturuben:~/kubernetes-aws-practice/app#
```

2.2 – Crear ConfigMap con archivos de la app *cat > configmap.yaml << 'EOF'*

```
apiVersion: v1
kind: ConfigMap
metadata:
 name: app-files
 namespace: load-balancer-demo
data:
 requirements.txt: |
 Flask==3.0.0
 Werkzeug==3.0.0


 app.py: |
 #!/usr/bin/env python3
 from flask import Flask, jsonify, send_from_directory
import os
 import socket
 from datetime import datetime
 import sys
 app = Flask(__name__)
 POD_NAME = os.getenv('POD_NAME', 'Unknown Pod')
POD_NAMESPACE = os.getenv('POD_NAMESPACE',
'default')  @app.route('/')
 def index():
 return send_from_directory('.', 'index.html')
 @app.route('/pod-info')
 def pod_info():
 return jsonify({
 'pod_name': POD_NAME,
 'namespace': POD_NAMESPACE,
 'hostname': socket.gethostname(),
 'timestamp': datetime.now().isoformat()
 })
```

```
@app.route('/health')
def health():
return jsonify({'status': 'healthy', 'pod': POD_NAME}), 200
if __name__ == '__main__':
print(f"[{POD_NAME}] Iniciando servidor Flask...", file=sys.stderr)
app.run(host='0.0.0.0', port=5000, debug=False)


index.html: |
<!DOCTYPE html>
<html>
<head>
<title>Kubernetes Load Balancer</title>
<style>
body {
font-family: Arial, sans-serif;
display: flex;
justify-content: center;
align-items: center;
min-height: 100vh;
margin: 0;
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  }
.container {
background: white;
padding: 50px;
border-radius: 10px;
box-shadow: 0 10px 25px rgba(0,0,0,0.2);
text-align: center;
max-width: 500px;
}
h1 {
color: #667eea;
```

```
margin: 0 0 30px 0;
}
.info {
background: #f0f0f0;
padding: 20px;
border-radius: 5px;
margin: 20px 0;
}
.pod-name {
font-size: 28px;
color: #764ba2;
font-weight: bold;
font-family: monospace;
}
.timestamp {
color: #666;
font-size: 13px;
margin-top: 10px;
}
.instruction {
background: #e0f2fe;
padding: 15px;
border-radius: 5px;
color: #0369a1;
font-size: 14px;
margin-top: 20px;
}
.refresh-button {
margin-top: 20px;
padding: 10px 20px;
background: #667eea;
```

```
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
font-size: 14px;
}
.refresh-button:hover {
background: #764ba2;
}
</style>
</head>
<body>
<div class="container">
<h1>Kubernetes Load Balancer</h1>
<div class="info">
<p style="margin: 0 0 10px 0; color: #666;">Pod atendiendo solicitud:</p> <p
class="pod-name" id="pod-name">Cargando...</p>
<p class="timestamp" id="timestamp"></p>
</div>
<div class="instruction">
Actualiza constantemente esta página para ver el <b>balanceo de carga en acción</b>
</div>
<button class="refresh-button" onclick="location.reload()">Actualizar Ahora</button>
</div>
<script>
fetch('/pod-info')
.then(r => r.json())
.then(data => {
document.getElementById('pod-name').textContent = data.pod_name; const
date = new Date(data.timestamp);
```

```
 document.getElementById('timestamp').textContent =
date.toLocaleString('es-ES');
 })
 .catch(e => {
 document.getElementById('pod-name').textContent = ' Error';
console.error('Error:', e);
 });
 </script>
 </body>
 </html>
 EOF


# Aplicar
kubectl apply -f configmap.yaml
# Verificar
kubectl get configmap -n load-balancer-demo
```

```
kind: ConfigMap
metadata:
 name: app-files
 namespace: load-balancer-demo
data:
 requirements.txt: |
 Flask==3.0.0
 Werkzeug==3.0.0

 app.py: |
 #!/usr/bin/env python3
 from flask import Flask, jsonify, send_from_directory  import os
 import socket
 from datetime import datetime
 import sys
 app = Flask(__name__)
 POD_NAME = os.getenv('POD_NAME', 'Unknown Pod')  POD_NAMESPACE = os.getenv('POD_NAMESP
ACE', 'default')  @app.route('/')
 def index():
 return send_from_directory('.', 'index.html')
 @app.route('/pod-info')
 def pod_info():
 return jsonify({
 'pod_name': POD_NAME,
 'namespace': POD_NAMESPACE,
 'hostname': socket.gethostname(),
 'timestamp': datetime.now().isoformat()
 })
 @app.route('/health')
 def health():
 return jsonify({'status': 'healthy', 'pod': POD_NAME}), 200  if __name__ == '__main__'
:
 print(f"[{POD_NAME}] Iniciando servidor Flask...", file=sys.stderr)  app.run(host='0.0
.0.0', port=5000, debug=False)
 index.html: |
 <!DOCTYPE html>
 <html>
kubectl get configmap -n load-balancer-demo tent = ' Error';  console.error('Error:', e
>
```

2.3 – Crear Deployment con 3 replicas

cat > deployment.yaml << 'EOF'

apiVersion: apps/v1

kind: Deployment

metadata:

name: web-app

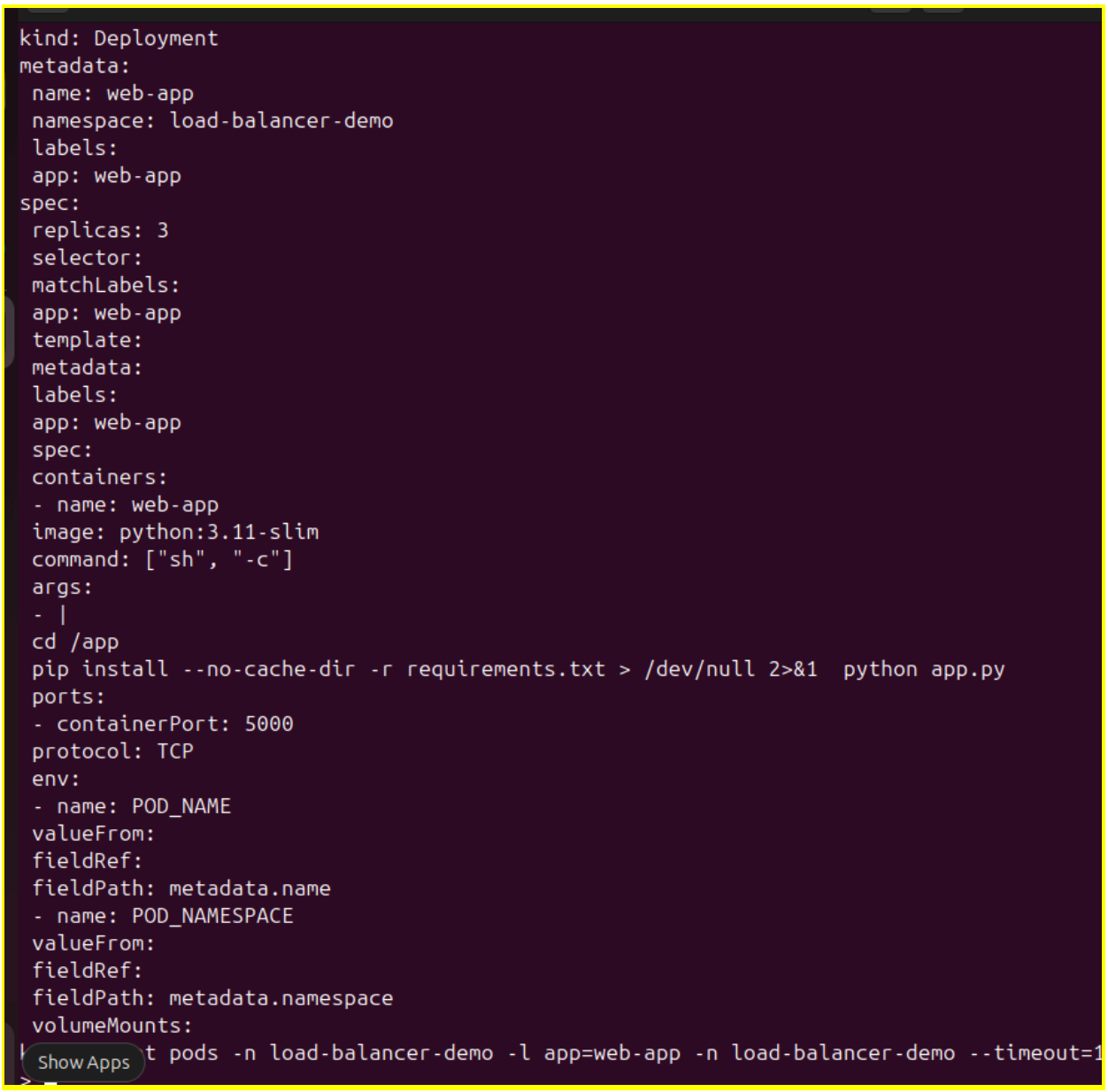
namespace: load-balancer-demo

labels:

app: web-app

```
spec:
 replicas: 3
 selector:
 matchLabels:
 app: web-app
 template:
 metadata:
 labels:
 app: web-app
 spec:
 containers:
 - name: web-app
 image: python:3.11-slim
 command: ["sh", "-c"]
 args:
 - |
 cd /app
 pip install --no-cache-dir -r requirements.txt > /dev/null 2>&1
python app.py
 ports:
 - containerPort: 5000
 protocol: TCP
 env:
 - name: POD_NAME
 valueFrom:
 fieldRef:
 fieldPath: metadata.name
 - name: POD_NAMESPACE
 valueFrom:
 fieldRef:
 fieldPath: metadata.namespace
```

```
 volumeMounts:

 - name: app-volume
 mountPath: /app

 livenessProbe:

 httpGet:

 path: /health

 port: 5000

 initialDelaySeconds: 15

 periodSeconds: 10

 readinessProbe:

 httpGet:

 path: /health

 port: 5000

 initialDelaySeconds: 5

 periodSeconds: 5

 volumes:

 - name: app-volume

 configMap:

 name: app-files

 defaultMode: 0755

 EOF


# Aplicar deployment

kubectl apply -f deployment.yaml

# Verificar que los pods se están creando

kubectl get pods -n load-balancer-demo

# Esperar a que estén ready (puede tardar 30-60 segundos)

 echo "Esperando a que los pods estén listos..."

kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo

--timeout=120s # Verificar

kubectl get pods -n load-balancer-demo
```



2.4 – Crear Service (*Load Balancer*)

cat > service.yaml << 'EOF'

apiVersion: v1

kind: Service

metadata:

name: web-app-service

namespace: load-balancer-demo

labels:

app: web-app

spec:

```
  type: LoadBalancer
  selector:
    app: web-app
  ports:
  - protocol: TCP
    port: 80
    targetPort: 5000
    name: http
  sessionAffinity: None
EOF

# Aplicar service
kubectl apply -f service.yaml
# Verificar el service
kubectl get svc -n load-balancer-demo
# Obtener IP del service
kubectl get svc -n load-balancer-demo web-app-service -o wide
```

```
 env:
 - name: POD_NAME
 valueFrom:
 fieldRef:
 fieldPath: metadata.name
 - name: POD_NAMESPACE
 valueFrom:
 fieldRef:
 fieldPath: metadata.namespace
 volumeMounts:
kubectl get pods -n load-balancer-demo -l app=web-app -n load-bala
> cat > service.yaml << 'EOF'
apiVersion: v1
kind: Service
metadata:
 name: web-app-service
 namespace: load-balancer-demo
 labels:
 app: web-app
spec:
 type: LoadBalancer
 selector:
 app: web-app
 ports:
 - protocol: TCP
 port: 80
 targetPort: 5000
 name: http
 sessionAffinity: None
EOF
# Aplicar service
kubectl apply -f service.yaml
# Verificar el service
kubectl get svc -n load-balancer-demo
# Obtener IP del service
kubectl get svc -n load-balancer-demo web-app-service -o wide
```

PARTE 3: VERIFICAR BALANCEO DE CARGA

LOCAL 3.1 – Levantar la aplicación

 # Opción 1: Port-forward (recomendado para WSL2)

kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80

```
kubectl get svc -n load-balancer-demo web-app-service -o wide
> # Opción 1: Port-forward (recomendado para WSL2)
kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80
```

3.2 – Probar balanceo con curl

 # Script para hacer peticiones y ver qué pod responde

for i in {1..10}; do

echo "Petición $i:"

curl -s http://localhost:8080/pod-info | python3 -m json.tool | grep

pod_name sleep 1

done

 # Deberías ver el mismo pod respondiendo varias veces lo siguiente:

 # "pod_name": "web-app-xxxxx-11111"

```
EOF
# Aplicar service
kubectl apply -f service.yaml
# Verificar el service
kubectl get svc -n load-balancer-demo
# Obtener IP del service
kubectl get svc -n load-balancer-demo web-app-service -o wide
> # Opción 1: Port-forward (recomendado para WSL2)
kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80
> # Script para hacer peticiones y ver qué pod responde
for i in {1..10}; do
 echo "Petición $i:"
 curl -s http://localhost:8080/pod-info | python3 -m json.tool | grep pod_name  sleep 1

done
# Deberías ver el mismo pod respondiendo varias veces lo siguiente:
# "pod_name": "web-app-xxxxx-11111"
>
```

3.3 – Verificar en navegador

Abre http://localhost:8080 en tu navegador y actualiza varias veces. Verás que cambia el pod que atiende.

PARTE 4: CONFIGURACIÓN EN AWS

4.1 – Crear Security Group

En AWS Console:

1. **Accede a EC2:** Security Groups
2. Haz clic en "Create security group"
3. **Nombre:** kubernetes-aws-sg
4. **Descripción:** Security group para Kubernetes con AWS

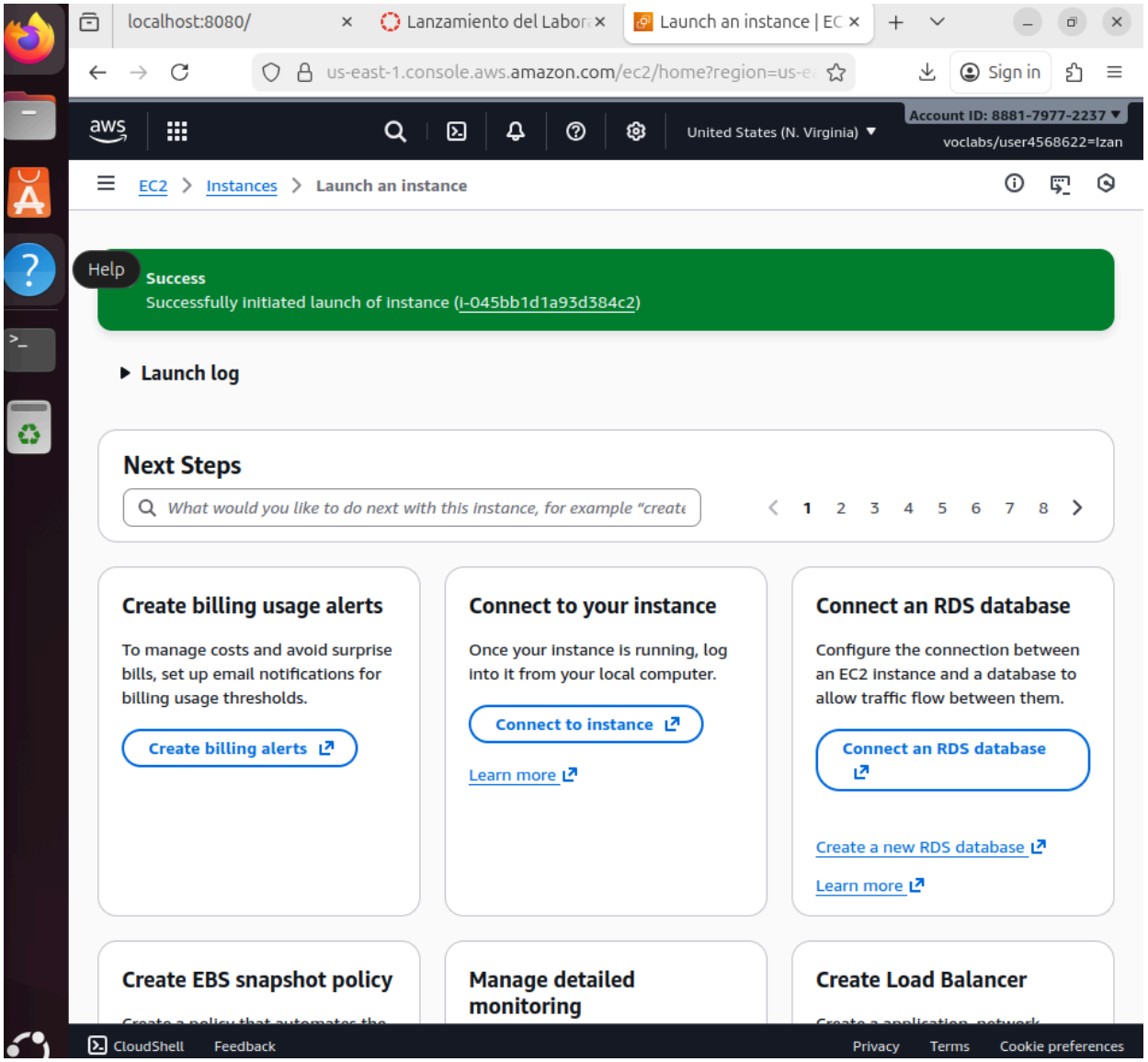
Agregar reglas de entrada:

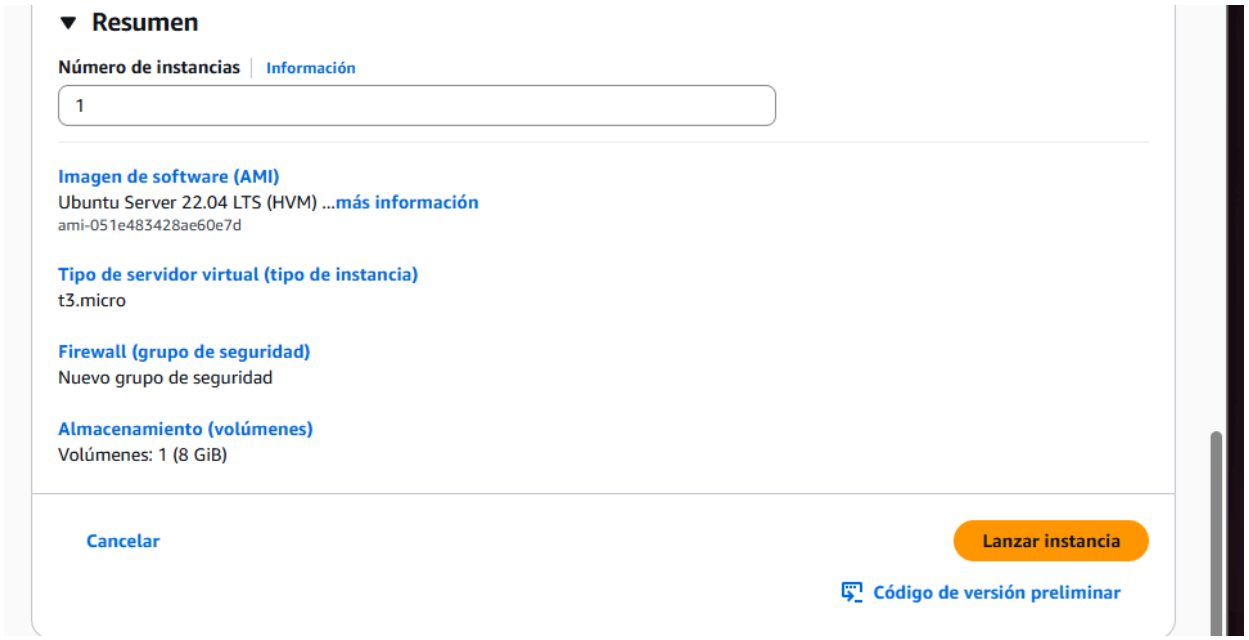
Tipo	Protocolo	Puerto	Origen
SSH	**TCP**	22	0.0.0.0/0
HTTP	**TCP**	80	0.0.0.0/0
HTTPS	**TCP**	443	0.0.0.0/0

4.2 – Crear instancia EC2

En AWS Console:

1. **EC2** → Instances → Launch instances
2. **Nombre:** kubernetes-test-server
3. **AMI:** Ubuntu 24.04 LTS
4. **Tipo:** t3.micro (*Free Tier*)
5. **Key pair:** Tu clave SSH
6. **VPC:** Default
7. **Security group:** kubernetes-aws-sg
8. **Storage:** 8 GiB, gp3
9. Launch instance


localhost:8080/
Lanzamiento del Labor
Launch an instance | EC
us-east-1.console.aws.amazon.com/ec2/home?region=us-e
Sign in
aws
United States (N. Virginia)
Account ID: 8881-7977-2237
voclabs/user4568622=Izan
EC2 > Instances > Launch an instance
Help
Success
Successfully initiated launch of instance (i-045bb1d1a93d384c2)
Launch log
Next Steps
What would you like to do next with this instance, for example "create
1 2 3 4 5 6 7 8
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Manage detailed monitoring
Create Load Balancer
CloudShell
Feedback
Privacy
Terms
Cookie preferences


Resumen
Número de instancias | Información
1
Imagen de software (AMI)
Ubuntu Server 22.04 LTS (HVM) ...más información
ami-051e483428ae60e7d
Tipo de servidor virtual (tipo de instancia)
t3.micro
Firewall (grupo de seguridad)
Nuevo grupo de seguridad
Almacenamiento (volúmenes)
Volúmenes: 1 (8 GiB)
Cancelar
Lanzar instancia
Código de versión preliminar

4.3 – Conectar a EC2

 # Obtener IP pública desde AWS Console (ej: 54.123.45.67)

 # Conectar via SSH

ssh -i ~/.ssh/clave-ssh.pem ubuntu@54.81.168.176

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '54.81.168.176' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Jan  9 10:30:32 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   25.8% of 6.71GB   Processes:             110
  Memory usage: 23%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.21.208

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

 # En EC2, instalar herramientas

sudo apt update

sudo apt install -y curl wget python3 python3-pip git

```
sudo apt install -y curl wget python3 python3-pip git
Get:1 https://dl.cloudsmith.io/public/caddy/stable/deb/debian any-version InRelease [14
.8 kB]
Hit:2 http://es.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 14.8 kB in 1s (28.0 kB/s)
Reading package lists... 78%
```

 # Crear directorio de trabajo

mkdir -p ~/kubernetes-test

```
root@ubunturuben:/home/izan# mkdir -p ~/kubernetes-test
root@ubunturuben:/home/izan#
```

Help

PARTE 5: CONECTAR KUBERNETES LOCAL CON AWS EC2

5.1 – Crear túnel SSH que expone el LoadBalancer

En una nueva máquina local (*WSL2*), mantén esta terminal abierta, el túnel estará activo mientras esté abierta:

 # Reemplaza 54.123.45.67 con tu IP de EC2

ssh -i ~/.ssh/tu-clave-aws.pem \ -N -R 8888:localhost:8080 \ ubuntu@54.123.45.67

 # -N: No ejecutar comandos remotos

 # -R 8888:localhost:8080: Redirige puerto 8888 en EC2 a puerto 8080 local

```
izan@ubunturuben:~$ ^[[200~ssh -i /home/izan/Downloads/clave-ssh.pem -N -R 8888:localho
st:8080 ubuntu@98.84.188.169
ssh: command not found
izan@ubunturuben:~$ ssh -i /home/izan/Downloads/clave-ssh.pem -N -R 8888:localhost:8080
 ubuntu@98.84.188.169
```

ubuntu@ip-172-31-21-208: ~

```
ubuntu@ip-172-31-21-208:~$ ss -tlnp | grep 8888
LISTEN 0      128        127.0.0.1:8888      0.0.0.0:*
LISTEN 0      128            [::1]:8888         [::]:*
ubuntu@ip-172-31-21-208:~$
```

Terminal

En AWS EC2 (*otra terminal local*):

 # Conecta a la instancia en otra terminal

ssh -i ~/.ssh/tu-clave-aws.pem ubuntu@54.123.45.67

 # Verificar que el túnel funciona 79

/

*curl -s http://localhost:8888/pod-info

```
ubuntu@ip-172-31-21-208:~$ curl -s http://localhost:8888/pod-info
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at localhost Port 8888</address>
</body></html>
```

 # Deberías recibir JSON:

 # {"pod_name":"web-app-xxxxx-11111", "namespace":"load-balancer-demo", ...}

5.2 – Crear script de prueba en EC2

En la instancia EC2:

```
cat > ~/test-kubernetes-lb.sh << 'EOF'

#!/bin/bash

echo "=== Prueba Kubernetes Load Balancer desde AWS ==="

echo ""

declare -A pods_count

for i in {1..15}; do

 response=$(curl -s http://localhost:8888/pod-info)

 pod_name=$(echo $response | python3 -c "import sys, json;
print(json.load(sys.stdin)['pod_name'])" 2>/dev/null)

 if [ -z "$pod_name" ]; then
 pod_name="ERROR"

 fi

 echo "Petición $i → Pod: $pod_name"

 pods_count[$pod_name]=$((${pods_count[$pod_name]:-0} + 1))

 sleep 1
```

done

echo ""

echo "=== Resumen Balanceo ==="

for pod in "${!pods_count[@]}"; do

echo "Pod $pod: ${pods_count[$pod]} peticiones"

done

EOF

sudo chmod +x ~/test-kubernetes-lb.sh

 # Ejecutar prueba

~/test-kubernetes-lb.sh

```
echo "=== Prueba Kubernetes Load Balancer desde AWS ==="
echo "" -A pods_count
declare -A pods_count
for i in {1..15}; do ttp://localhost:8888/pod-info)
 response=$(curl -s http://localhost:8888/pod-info) json;
 pod_name=$(echo $response | python3 -c "import sys, json;
print(json.load(sys.stdin)['pod_name'])" 2>/dev/null)
 if [ -z "$pod_name" ]; then
 pod_name="ERROR"
 fi o "Petición $i → Pod: $pod_name"
 echo "Petición $i → Pod: $pod_name" t[$pod_name]:-0} + 1))
 pods_count[$pod_name]=$((${pods_count[$pod_name]:-0} + 1))
 sleep 1
done ""
echo "" = Resumen Balanceo ==="
echo "=== Resumen Balanceo ===" do
for pod in "${!pods_count[@]}"; do } peticiones"
 echo "Pod $pod: ${pods_count[$pod]} peticiones"
done
EOF  chmod +x ~/test-kubernetes-lb.sh
sudo chmod +x ~/test-kubernetes-lb.sh
# Ejecutar prueba lb.sh
~/test-kubernetes-lb.sh
>
```

PARTE 6: ESCALADO DINÁMICO

6.1 – Escalar a 5 replicas

En tu máquina local:

 # Aumentar a 5 replicas

kubectl scale deployment web-app -n load-balancer-demo

--replicas=5 # Verificar

kubectl get pods -n load-balancer-demo

 # Esperar a que estén ready

kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo --timeout=120s

```
> kubectl scale deployment web-app -n load-balancer-demo --replicas=5 # Verifica
r
kubectl get pods -n load-balancer-demo
# Esperar a que estén ready
kubectl wait --for=condition=ready pod -l app=web-app -n load-balancer-demo --ti
meout=120s
> S
```

6.2 – Probar balanceo desde AWS

En EC2:

~/test-kubernetes-lb.sh

 # Hay más variedad de pods, pero solo va aparecer uno

```
ubuntu@ip-172-31-21-208:~$ bash ~/test-kubernetes-lb.sh
```

PARTE 7: MONITOREO Y OBSERVABILIDAD

7.1 – Ver estado de pods

 # Ver todos los pods

kubectl get pods -n load-balancer-demo

```
izan@ubunturuben:~$ kubectl get pods -n load-balancer-demo
NAME                          READY   STATUS    RESTARTS   AGE
web-app-c8b4f854b-8mjq2   1/1     Running   0          27s
web-app-c8b4f854b-nvxpz   1/1     Running   0          27s
web-app-c8b4f854b-phspj   1/1     Running   0          27s
izan@ubunturuben:~$
```

 # Ver logs de un pod específico

kubectl logs -n load-balancer-demo

```
izan@ubunturuben:~$ kubectl logs -n load-balancer-demo -l app=web-app --tail=10 -f
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/16 09:40:40 [notice] 1#1: start worker processes
2026/01/16 09:40:40 [notice] 1#1: start worker process 29
2026/01/16 09:40:40 [notice] 1#1: start worker process 30
2026/01/16 09:40:40 [notice] 1#1: start worker process 31
2026/01/16 09:40:40 [notice] 1#1: start worker process 32
2026/01/16 09:40:40 [notice] 1#1: start worker process 33
2026/01/16 09:40:40 [notice] 1#1: start worker process 34
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/16 09:40:40 [notice] 1#1: start worker processes
2026/01/16 09:40:40 [notice] 1#1: start worker process 30
2026/01/16 09:40:40 [notice] 1#1: start worker process 31
2026/01/16 09:40:40 [notice] 1#1: start worker process 32
2026/01/16 09:40:40 [notice] 1#1: start worker process 33
2026/01/16 09:40:40 [notice] 1#1: start worker process 34
2026/01/16 09:40:40 [notice] 1#1: start worker process 35
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
```

web-app-xxxxx-11111 # Ver logs en tiempo real

kubectl logs -n load-balancer-demo -l app=web-app -f

```
izan@ubunturuben:~$ kubectl logs -n load-balancer-demo -l app=web-app -f
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/16 09:40:40 [notice] 1#1: start worker processes
2026/01/16 09:40:40 [notice] 1#1: start worker process 29
2026/01/16 09:40:40 [notice] 1#1: start worker process 30
2026/01/16 09:40:40 [notice] 1#1: start worker process 31
2026/01/16 09:40:40 [notice] 1#1: start worker process 32
2026/01/16 09:40:40 [notice] 1#1: start worker process 33
2026/01/16 09:40:40 [notice] 1#1: start worker process 34
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/16 09:40:40 [notice] 1#1: start worker processes
2026/01/16 09:40:40 [notice] 1#1: start worker process 29
2026/01/16 09:40:40 [notice] 1#1: start worker process 30
2026/01/16 09:40:40 [notice] 1#1: start worker process 31
2026/01/16 09:40:40 [notice] 1#1: start worker process 32
2026/01/16 09:40:40 [notice] 1#1: start worker process 33
2026/01/16 09:40:40 [notice] 1#1: start worker process 34
2026/01/16 09:40:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/16 09:40:40 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/16 09:40:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/16 09:40:40 [notice] 1#1: start worker processes
```

7.2 – Ver estadísticas

 # CPU y memoria de pods

kubectl top pods -n load-balancer-demo

```
izan@ubunturuben:~$ kubectl top pods -n load-balancer-demo
NAME                      CPU(cores)   MEMORY(bytes)
web-app-c8b4f854b-8mjq2   0m           6Mi
web-app-c8b4f854b-nvxpz   0m           6Mi
web-app-c8b4f854b-phspj   0m           6Mi
izan@ubunturuben:~$
```

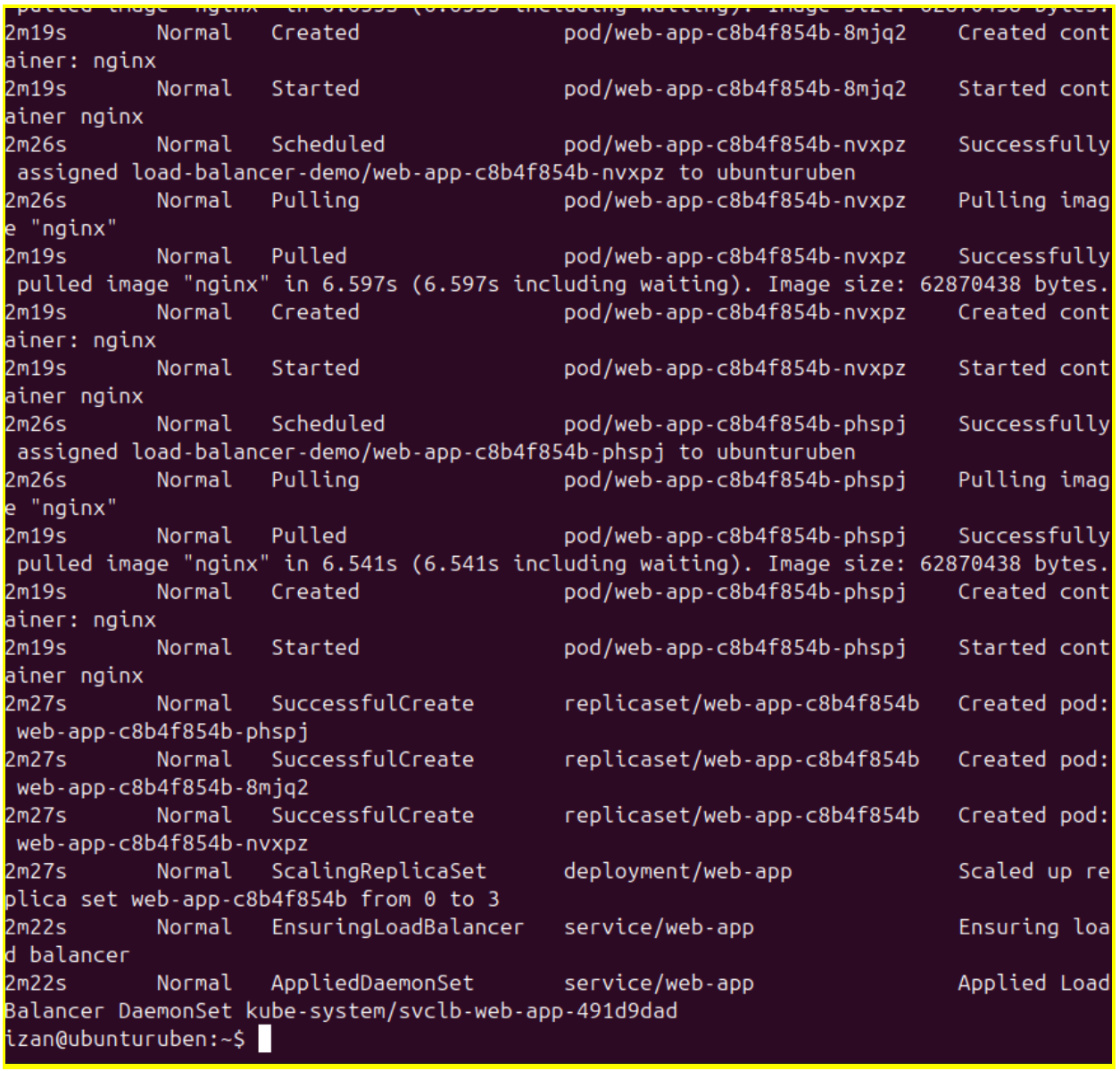
 # Nodos del cluster

kubectl top nodes

```
izan@ubunturuben:~$ kubectl top nodes
NAME          CPU(cores)   CPU(%)   MEMORY(bytes)   MEMORY(%)
ubunturuben   222m         3%       3106Mi          58%
izan@ubunturuben:~$
```

 # Eventos del cluster

kubectl get events -n load-balancer-demo



7.3 – Ver detalles del deployment

 # Información completa del deployment

kubectl describe deployment web-app -n

```
izan@ubunturuben:~$ kubectl describe deployment web-app -n load-balancer-demo
Name:                   web-app
Namespace:              load-balancer-demo
CreationTimestamp:      Fri, 16 Jan 2026 09:40:32 +0000
Labels:                 app=web-app
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=web-app
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Terminal
  Labels:  app=web-app
  Containers:
   nginx:
    Image:         nginx
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   web-app-c8b4f854b (3/3 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  3m12s  deployment-controller  Scaled up replica set web-ap
p-c8b4f854b from 0 to 3
```

load-balancer-demo # Historial de cambios

kubectl rollout history deployment web-app -n load-balancer-demo

```
izan@ubunturuben:~$ kubectl rollout history deployment web-app -n load-balancer-demo
deployment.apps/web-app
REVISION  CHANGE-CAUSE
1         <none>

izan@ubunturuben:~$
```

CTRL DERECHA

PARTE 8: ELIMINAR RECURSOS

8.1 – Limpiar Kubernetes

 # Eliminar todo en el namespace

kubectl delete namespace load-balancer-demo

```
izan@ubunturuben:~$ kubectl delete namespace load-balancer-demo
namespace "load-balancer-demo" deleted
```

 # Verificar

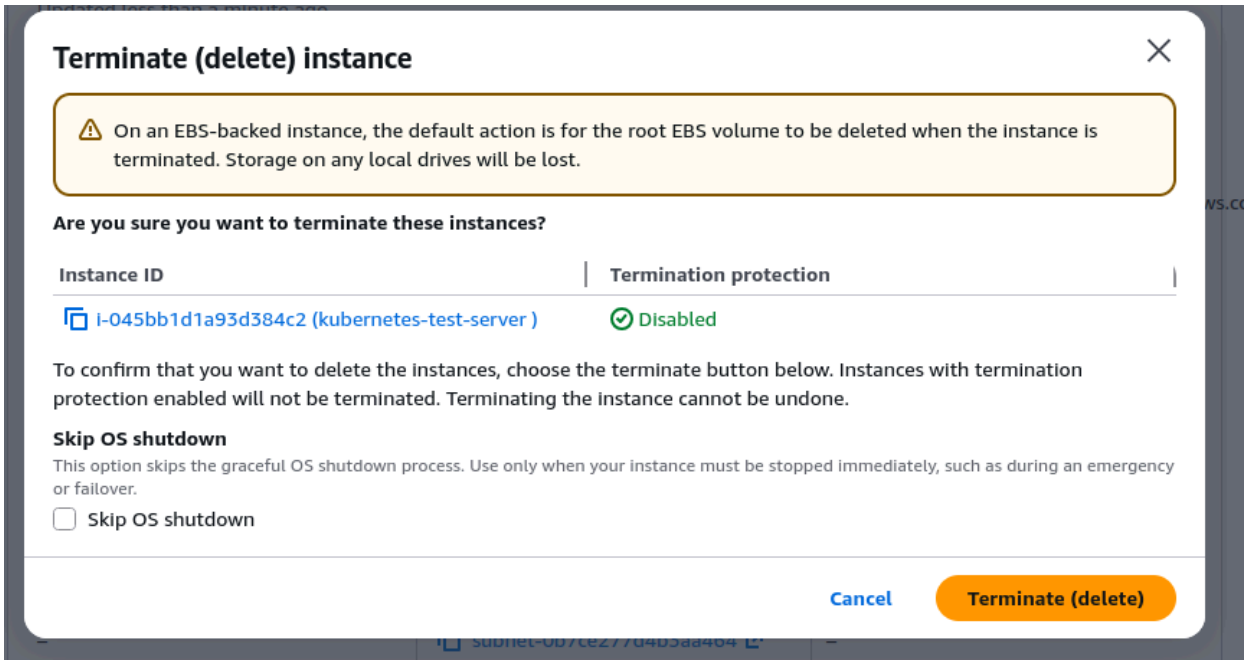
kubectl get namespaces

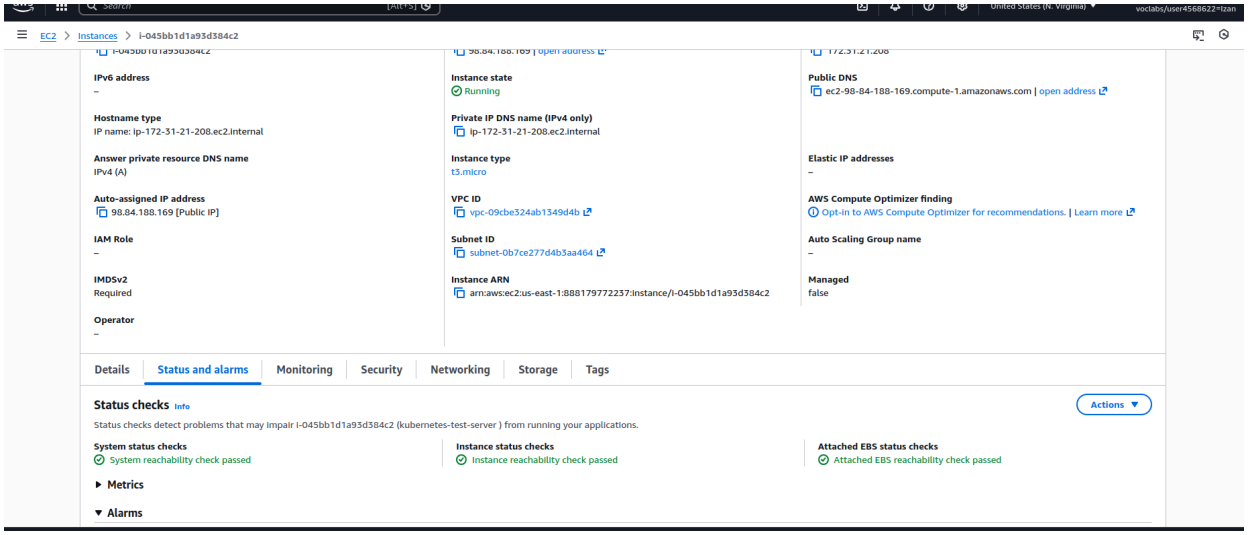
```
izan@ubunturuben:~$ kubectl delete namespace load-balancer-demo
namespace "load-balancer-demo" deleted
^Z
[3]+  Stopped                 kubectl delete namespace load-balancer-demo
izan@ubunturuben:~$ kubectl get namespaces
NAME                 STATUS        AGE
default              Active        7d
kube-node-lease      Active        7d
kube-public          Active        7d
kube-system          Active        7d
load-balancer-demo   Terminating   3m53s
izan@ubunturuben:~$
```

8.2 – Eliminar instancia EC2

En AWS Console:

1. **EC2** → Instances
2. Selecciona kubernetes-test-server
3. **Instance State** → Terminate
4. Confirma





8.3 – Detener k3s

 # Si quieres pausar k3s

sudo systemctl stop k3s

 # Para eliminar completamente

sudo /usr/local/bin/k3s-uninstall.sh

```
izan@ubunturuben:~$ sudo systemctl stop k3s
[sudo] password for izan:
izan@ubunturuben:~$ sudo /usr/local/bin/k3s-uninstall.sh
+ id -u
+ [ 0 -eq 0 ]
+ K3S_DATA_DIR=/var/lib/rancher/k3s
+ /usr/local/bin/k3s-killall.sh
+ [ -s /etc/systemd/system/k3s.service ]
+ basename /etc/systemd/system/k3s.service
+ systemctl stop k3s.service
+ [ -x /etc/init.d/k3s* ]
+ killtree 4113 4148 4181 4195 4258
+ kill -9 4113 4242 4485 4148 4273 4470 4181 4276 4434 4195 4292 4438 4258 4313
4478 4629
+ do_unmount_and_remove /run/k3s
```

NOTAS IMPORTANTES

- **k3s vs Minikube:** k3s es más ligero y no necesita Docker ni drivers. Perfecto para WSL2.
- **Sin Docker:** Esta práctica NO usa Docker en ningún momento. Solo

Python puro.

- **Túnel SSH:** Es la forma más simple conectar local con AWS sin complicaciones de networking.
- **Balanceo real:** Kubernetes distribuye tráfico entre pods. Ver cambios en el navegador.
- **Costos:** t3.micro está en Free Tier. Elimina cuando termines.
- **Datos:** Los datos no persisten entre reinicios de pods (*sin BD*).

TROUBLESHOOTING

Error: "Unable to pick a default driver"

- k3s está instalado. Si aparece este error, ignóralo. k3s no necesita driver. **Error: "Connection refused" desde EC2**
- El túnel SSH no está activo
- **Solución:** Verificar sesión SSH con -R en terminal local **Pods en "Pending"**
- k3s no tiene suficientes recursos o está iniciando

kubectl describe pod <pod-name> -n load-balancer-demo

Port-forward no responde

 # Matar procesos previos

pkill -f "port-forward"

 # Reiniciar

kubectl port-forward -n load-balancer-demo svc/web-app-service 8080:80 **ConfigMap no se actualiza**

- Los pods en ejecución mantienen versión anterior

kubectl rollout restart deployment web-app -n load-balancer-demo

- k3s no inicia después de actualizar

sudo /usr/local/bin/k3s-uninstall.sh

curl -sfL https://get.k3s.io | K3S_KUBECONFIG_MODE="644" sh -